Model Assessment & Adjusted R^2 Transcript

Hi everyone, in our last lesson, we successfully built a multiple linear regression model. Now it's time to become more critical about the kind of model that we have built. So in this lesson, we'll explore advanced topics that separate a basic model from a robust model, or we can say from a basic model to a more reliable model.

The idea is to create a model with not all the variables which we have included initially, to only have the variables which actually affect the target variable. So we will get introduced to terminologies like adjusted R square, which gets to our helper, which we use for understanding which particular set of variables or which variables exactly which variables are affecting the target variable, or actually making sense for the entire model. Now we move into how do we assess the model that we have got, and how do we choose the best model.

Okay, for that, we have already learned, we have a metric which is called as R square, R square tells us the amount of variation in y explained by the input variable or the x variables. That's what we have learned in simple linear regression. Now the problem with R square is in the case of multiple linear regression, as you keep on increasing more and more or adding more and more variable, let's say we start with in the previous example, we started with this data, right? So let's say we keep on adding more dimensions or more variables.

Now we are not only limited to bedrooms and square feet, we might want to add location, we might want to add whether it has swimming pool or not, we might want to add whether it has a garden x, y, z, a lot of more input variables, right? Now as we keep adding more and more variables, what happens to R square, R square, let's say if we add a random predictor or an or any predictor, the R square value would keep on increasing. That's a problem of R square. At every click which I am doing here, you see that the value of R square gets going or increasing towards one, right? This is actually the problem with R square.

Now to deal with this problem, we have invented a new thing which is called as adjusted R square. Adjusted R square, they only increase if the new predictor improves the model. What do we mean by improves the model? It means that, let's say for example, if I insert here, let's say whether it has swimming pool or not, I have added a new dimension to it, which says whether it has swimming pool.

One says it has swimming pool, zero says it does not have, okay? Now let's say this is our data. Now when we started, when we started with, might have started with the square fit and then we added bedroom, our R square value would have increased, maybe it would have increased from some, let's say it might have increased from 0.6 to 0.7. When we added swimming pool to it, it might have increased from 0.7 to 0.75. The problem is as you keep on increasing more and more variable, R square value increases. Now R square is a metric which tells how much amount of variance in the output variable or the target variable or what we have, what we call as dependent variable, what proportion of that variation is coming from the input variables.

So we kind of add some sort of penalty to R square such that it only increases when we add up a variable which actually affects the price, okay? We do not add any random variable. We do not add anything which does not affect price. So R, adjusted R square would only increase if swimming pool actually affects the price of the house.

I hope this all makes sense to all of you. So standard R square, it goes up with every predictor being added. Adjusted R square would only go up if that predictor makes sense for the target variable.

So this is how it actually works. We have added a penalty and that penalty is described here in the top in the numerator term. R square value is nothing but 1 minus sum of squared of errors divided by total sum of squares, right? Now in this, what we do, we divide sum of squares of errors by n minus k minus 1. What is the effect of that? As you increase the number of predictors or k increases, the term n minus k minus 1 decreases, right? As you increase k, n minus k minus 1 would decrease which eventually would increase the entire fraction, right? And thus, the adjusted R square, it becomes smaller.

The new variable must cause a substantial decrease in sum of squared of errors. It means that unless and until the newly added k or newly added k plus 1, the new variable that you have added, if it doesn't affect sum of squared of errors, right? If it doesn't minimise the sum of squared of errors, which means that it is actually affecting the target variable, then the R square would never increase. If it affects, then sum of squared of errors would decrease and this particular thing would also decrease.

Hence, adjusted R square would increase. So when you have different, different, different models, let's say you have model 1 wherein you have the sales of a company which is only attributed to ad spent, your adjusted R square is 0.65. When you add promotions to it as well, your adjusted R square is 0.72. But when you add web traffic, which actually doesn't affect sales, right? Then your adjusted R square is actually going down. It is not increasing.

So you would have to be really cautious while adding new variables, let's say, or new independent variables like promotions or web traffic. If it actually increases R square or adjusted R square, then only that variable is meaningful or then only that independent variable is meaningful. Otherwise, that variable is not meaningful.

We should not be adding that. I hope this was clear to everyone and I hope each one of you is able to understand what is the implication of adding any random variable to our data.

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